

Percentage of quantities



- 1 Complete the table of fraction to percentage conversions.

Fraction	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{8}$		$\frac{1}{20}$		$\frac{1}{50}$	$\frac{1}{100}$
Percentage		50%				10%		4%		1%

- 2 Write the fraction or number that completes the following statements:

- a 25% of a quantity is the same as $\frac{\quad}{\quad}$ of that quantity.
 b 200% of a quantity is the same as $\frac{\quad}{\quad}$ of that quantity.

- 3 Find the following quantities by first converting the percentage into a fraction.

- a 15% of 160 b 250% of 12 c 16% of 275 d 175% of 20

- 4 Find the following quantities by first converting the percentage into decimal.

- a 90% of 20 b 220% of 10 c 60% of 60 d 21% of 400

- 5 Show a way to correctly calculate 25% of 140.

- 6 Find the following:

- a 10% of 90 b 20% of 45 c 1% of 900 d 2% of 400
 e 500% of 11 f 350% of 16 g 60% of 30 h 45% of 160
 i 200% of 13 j 150% of 18 k 80% of 40 l 50% of 320

- 7 Find the following:

- a 10% of 80 g b 20% of \$60 c 30% of \$200 d 2% of 300 mL
 e 90% of 80 kg f 70% of 20 km g 25% of 120 L h 75% of \$24

- 8 Follow the steps in each case to determine the given quantity:

- a 75% of 28.
 i How many groups of 25% go into 75%?
 ii What is 25% of 28?
 iii Hence, find 75% of 28.

b 140% of 35.



- i** How many groups of 20% go into 140%?
- ii** What is 20% of 35?
- iii** Hence, find 140% of 35.

c 28% of 175.

- i** How many groups of 4% go into 28%?
- ii** What is 4% of 175?
- iii** Hence, find 28% of 175.

d 7% of 300.

- i** How many groups of 1% go into 7%?
- ii** What is 1% of 300?
- iii** Hence, find 7% of 300.

Percentage comparisons

9 Which is greater:

- | | | | | | |
|--------------------|----|-------------|---------------------|----|-------------|
| a 62% of 38 | or | 62% of 46? | b 27% of 230 | or | 13% of 230? |
| c 70% of 30 | or | 250% of 18? | d 45% of 180 | or | 7% of 700? |
| e 13% of 17 | or | 29% of 13? | f 36% of 48 | or | 24% of 63? |

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- a** When finding a percentage of 82 that is above 100%, will the result be smaller or larger than 82?
- b** When finding a percentage of 82 that is below 100%, will the result be smaller or larger than 82?
- c** Find the following:
 - i** 60% of 82
 - ii** 120% of 82

Applications

11 Lisa scored 95% on her geography exam, which was marked out of 160. What was Lisa's actual mark out of 160?

12 A theatre has a capacity of 3000 people. On opening night the theatre was 97% full. On the second night it was 78% full.



- a** How many people attended the theatre on the first night?
- b** How many people attended theatre second night?
- c** How many people attended the theatre in total during the first 2 nights.

13 A stadium can hold 5000 people. The stadium is 70% full and 11% of the audience are children.

- a** How many people are there in the stadium?
- b** How many children are there in the stadium?

14 Yuri is eating a box of cookies. The nutritional information states that each cookie contains 7% of his recommended daily intake of sugar.

If his recommended daily intake of sugar is 37 500 mg, how many milligrams of sugar does each cookie contain?